Inteligencia Artificial

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Clase # 18 - PAO 2026 – II

Agenda

1. Transformers
    1.1 Qué son ?
    1.2 Encoder – Decoder
    1.3 Auto-atenttion
    1.4 Las matrices: K, Q, V
    1.5 Multi-heads

Transformers

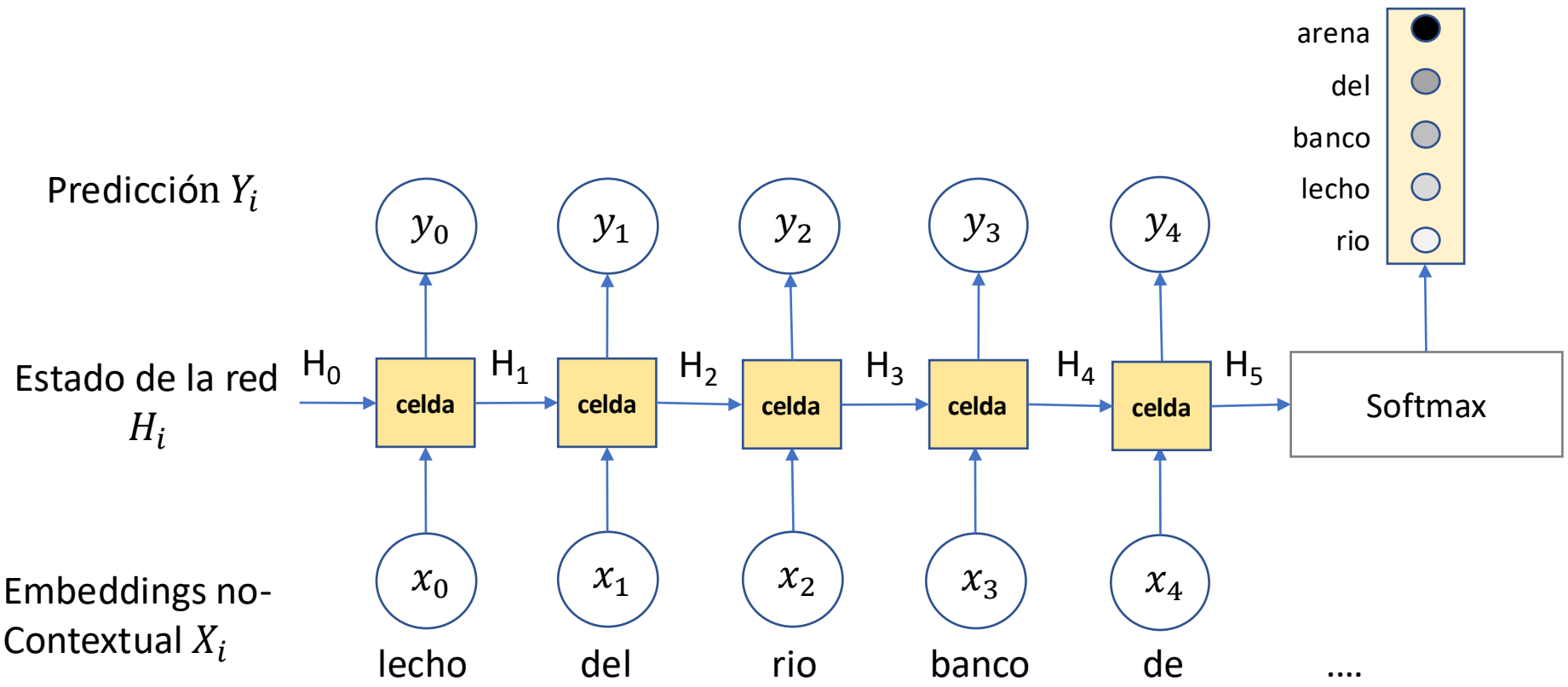
- 2013 – Word2Vec
- 2014 – RNNs modelos de lenguaje
- 2017 - Transformers

Transformers:

- Estándar de rendimiento en una variedad de tareas de NLP
- BERT (Bidirectional Encoder Representation Transformer),
- GPT (Generative Pre-trained Transformer) y otros,
- Estos modelos son capaces de captar sutilezas del lenguaje y generar texto

Transformers

- En las RNN

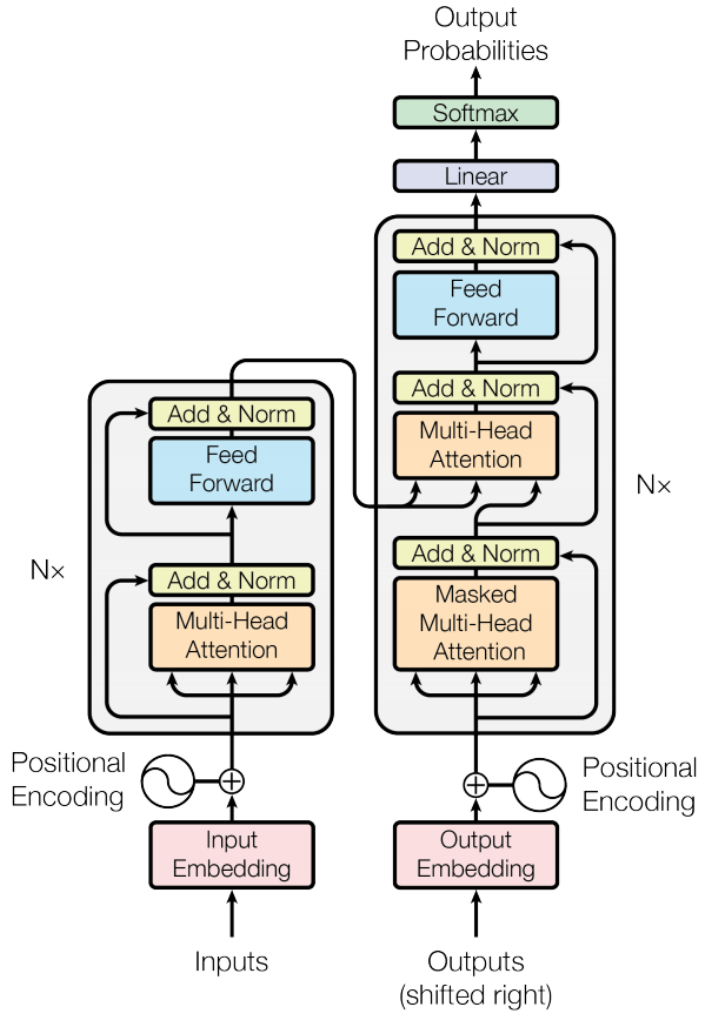


Retos:

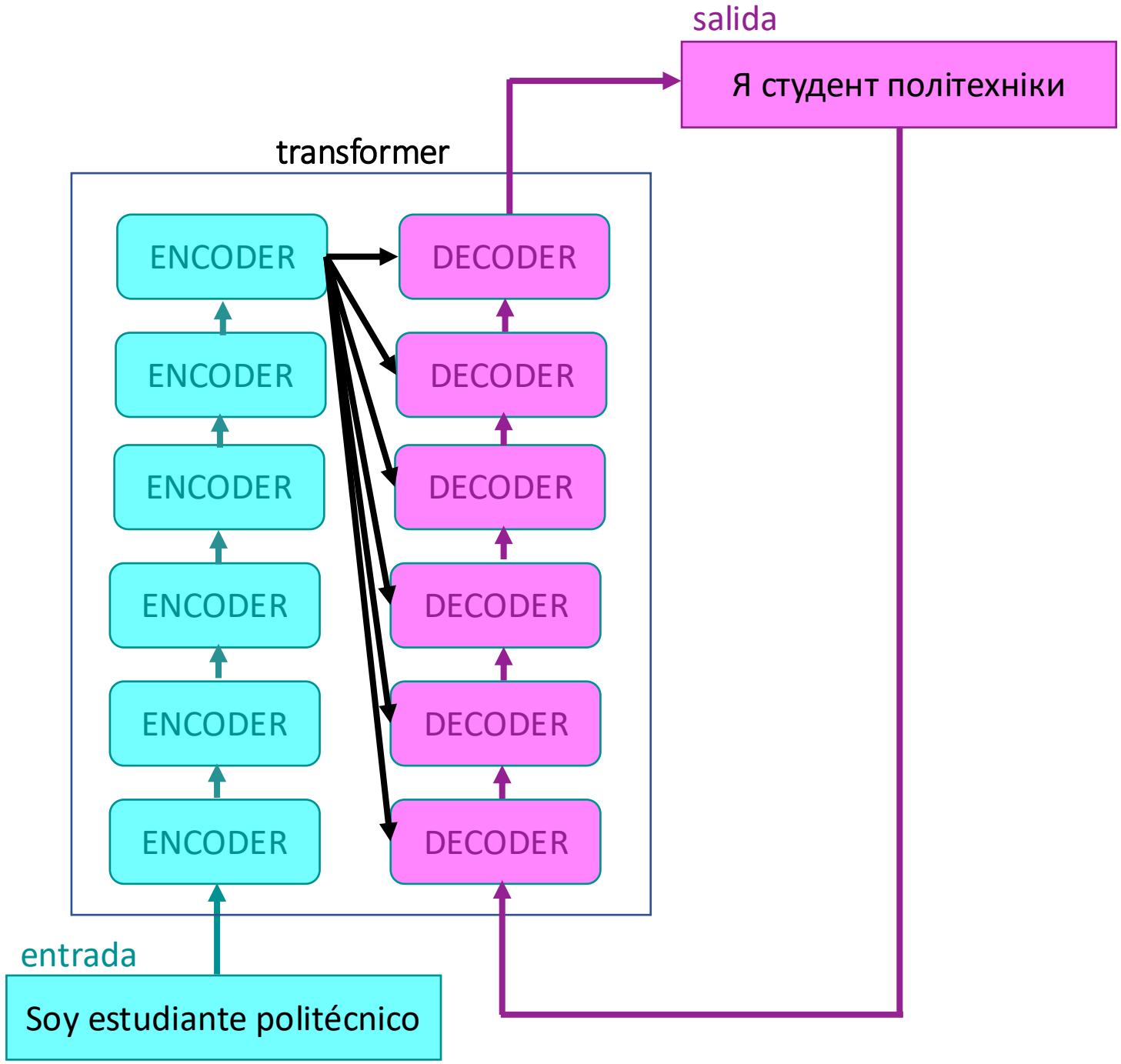
- Lento O(N): N # de tokens
- Desvanecimiento del gradiente
- Unidireccional: Izquierda – derecha
  - Banco de arena vs.
  - Arena en el banco

Transformers – Qué son ?

- Es un tipo de RN para tareas tipo Seq2Seq.
- Arquitectura encoder-decoder.
- Sin desvanecimiento del gradiente.
- Sin problema con las dependencias largas.
- Utilizan **attention** para aumentarla precisión y velocidad de entrenamiento.
- Propuesto en “Attention is all you need”.
- Su mayor beneficio, la paralelización.
- Diponible en TensorFlow como parte de Tensor2Tensor.
- También está en PyTorch.

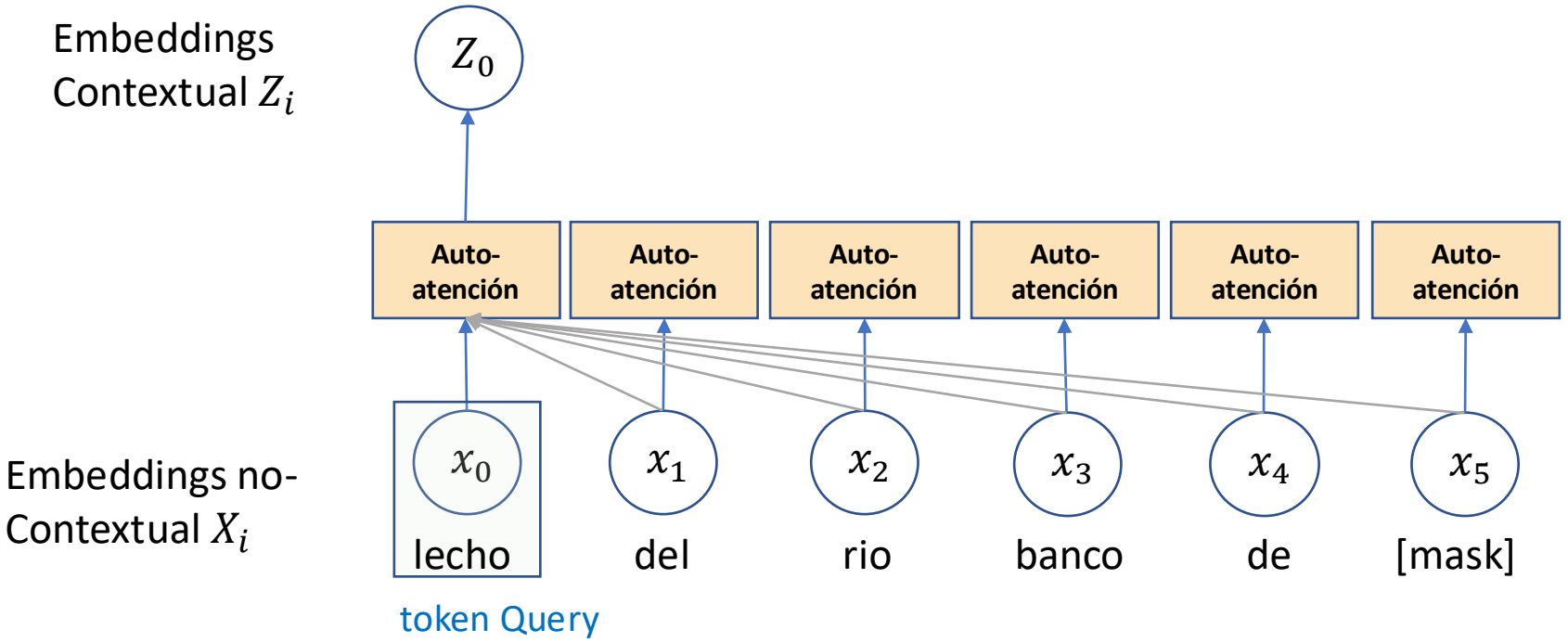


Transformers



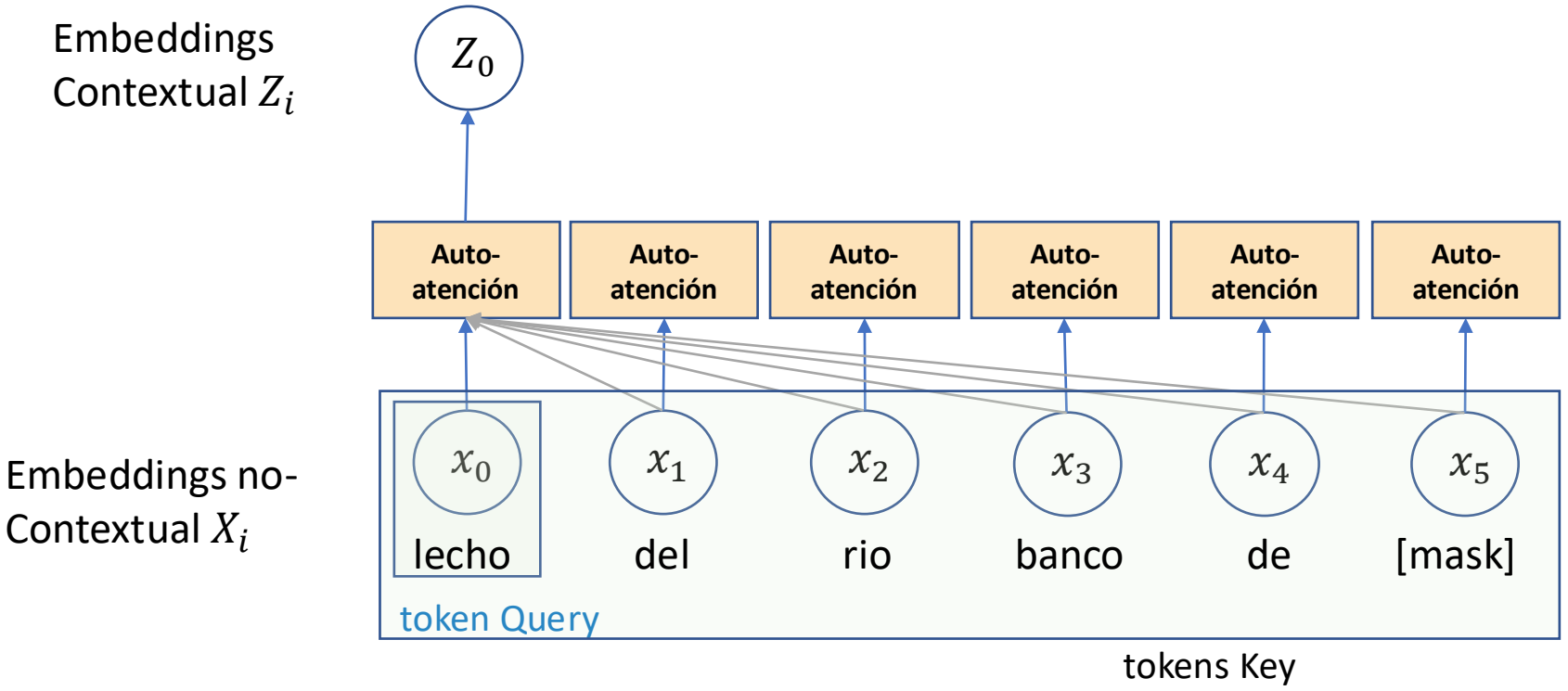
Transformers

- Auto-atención



Transformers

- Auto-atención

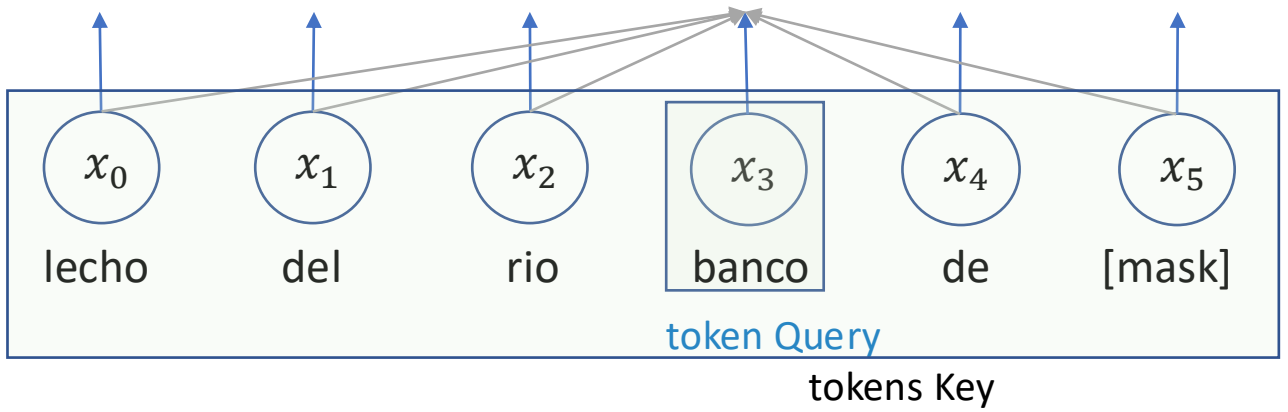


Transformers

- Los Valores (Values)

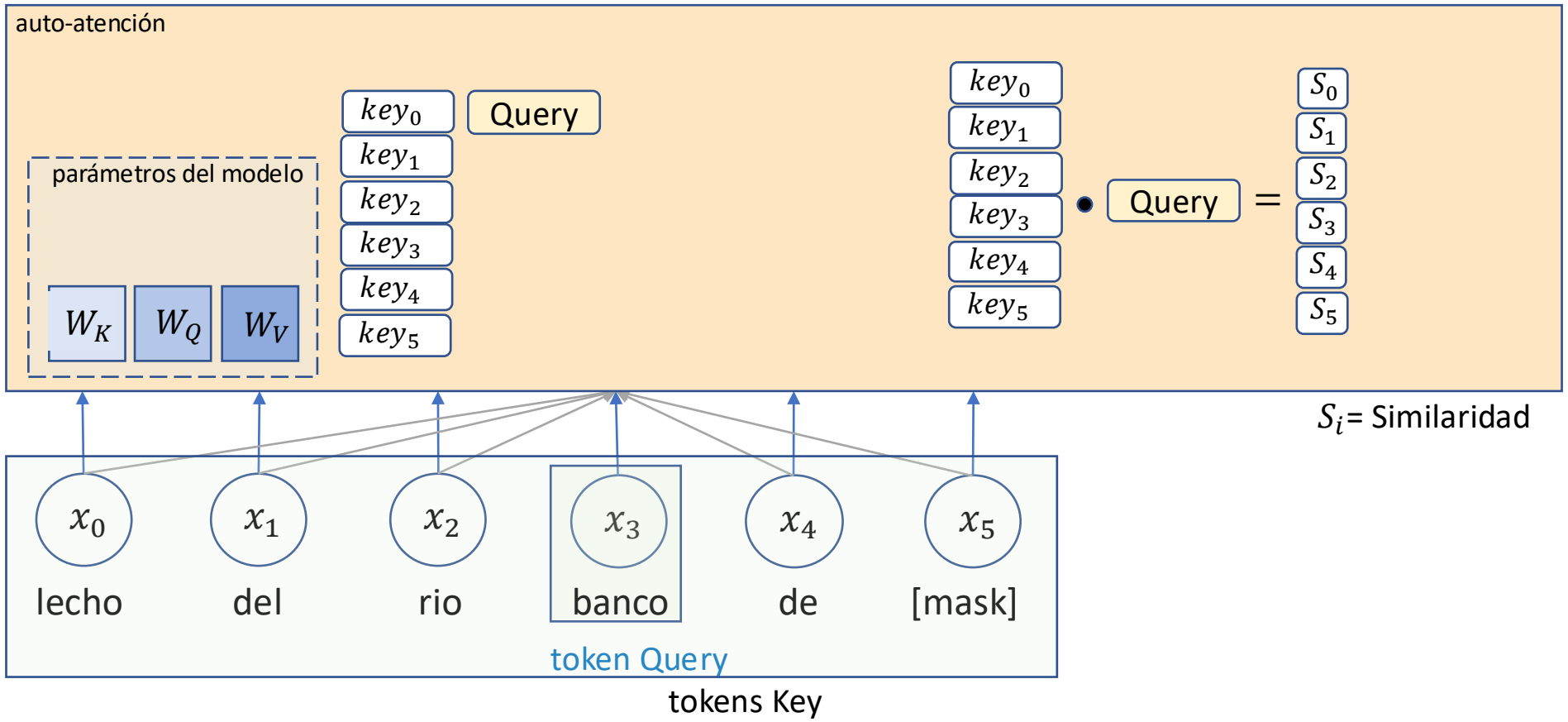
- Una analogía de Q, K y V podría ser Netflix:
  - Query: la búsqueda que escribe el usuario, por ejemplo: "películas de ciencia ficción".
  - Key: la información que Netflix usa para indexar cada película (género, actores, director, palabras clave, año, idioma, etc.). Es como la ficha técnica que permite encontrarla.
  - Value: el contenido que realmente se entrega cuando eliges esa película: el video, el audio, los subtítulos y toda la información asociada.

Embeddings no-Contextual X_i



Transformers

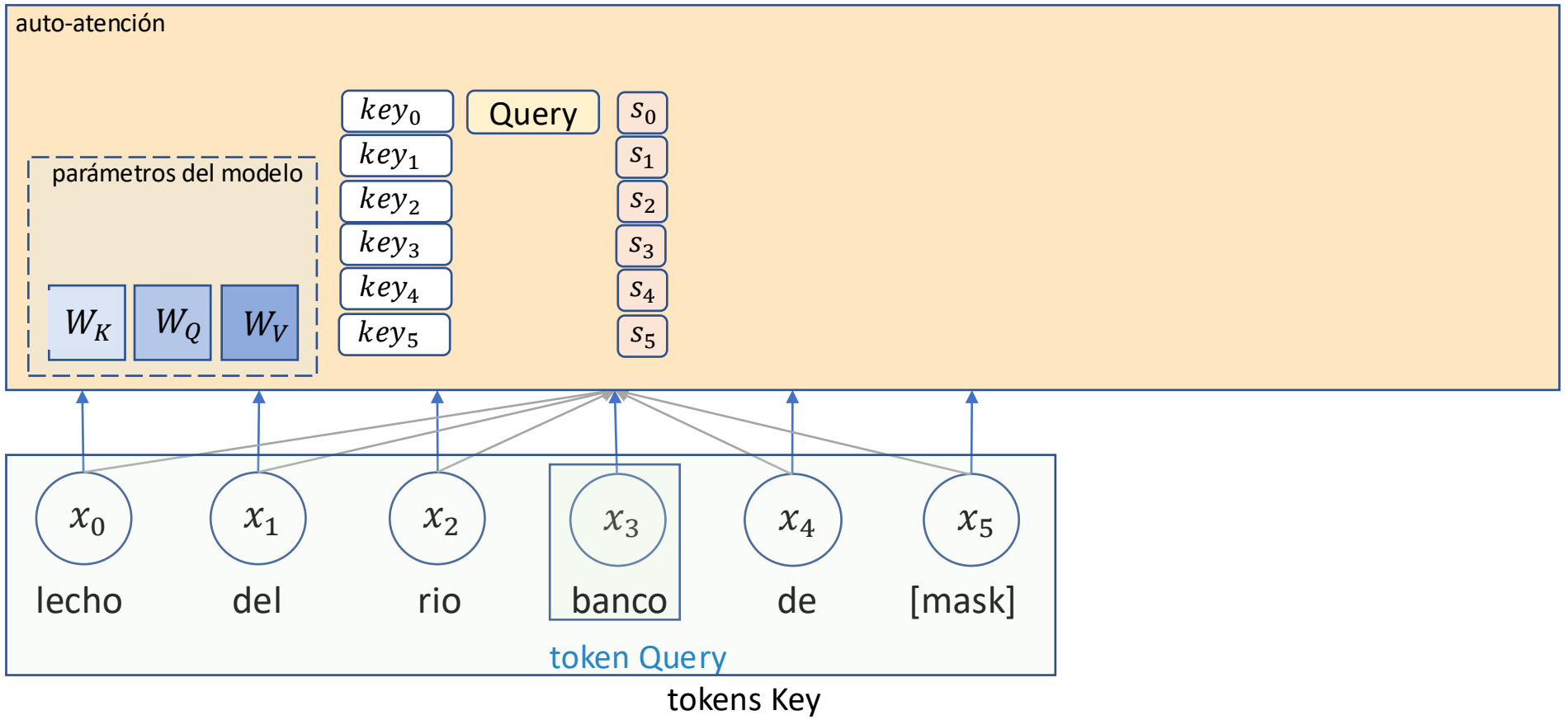
- Similitud



Embeddings no-Contextual X_i

Transformers

- Similitud

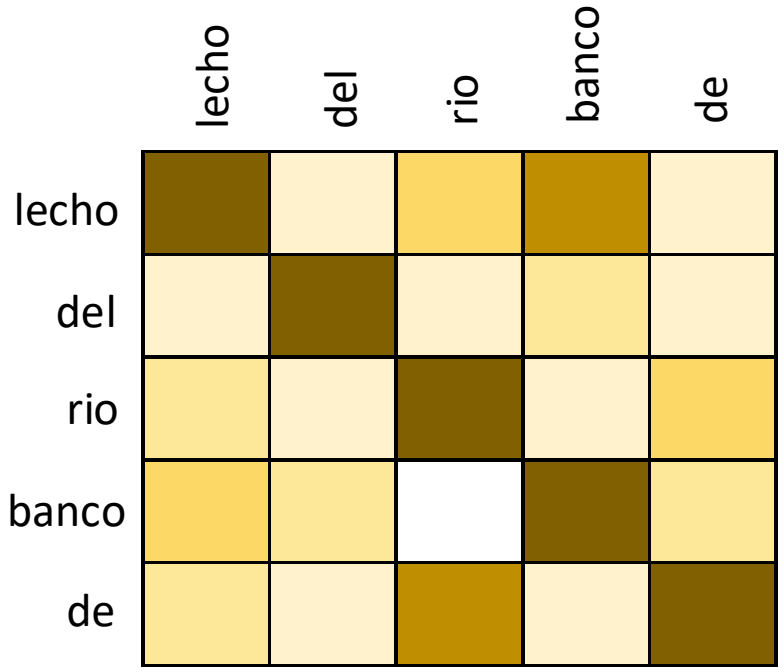


Embeddings no-Contextual X_i

Transformers

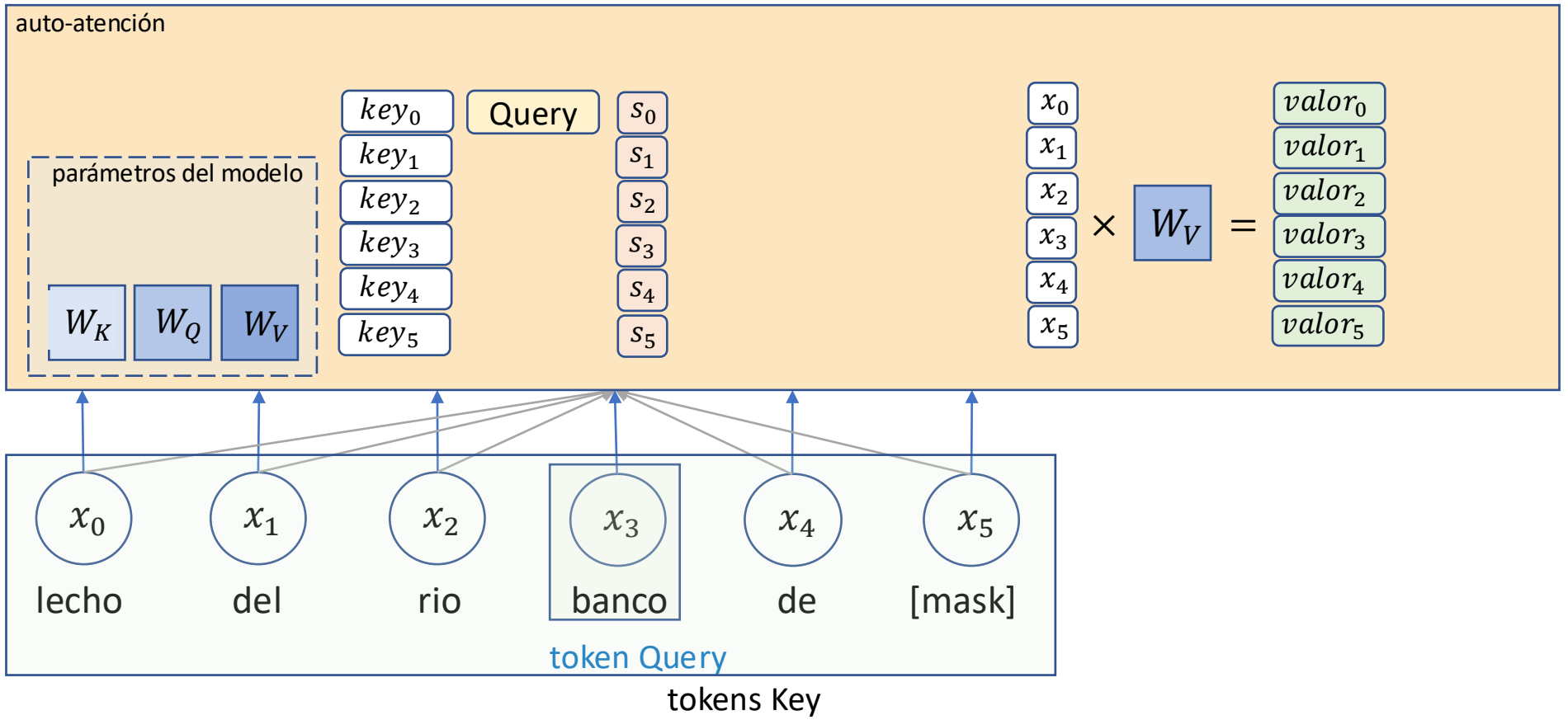
- Representa cuán similar es el Query con respecto a las Claves

$$softmax\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right)$$



Transformers

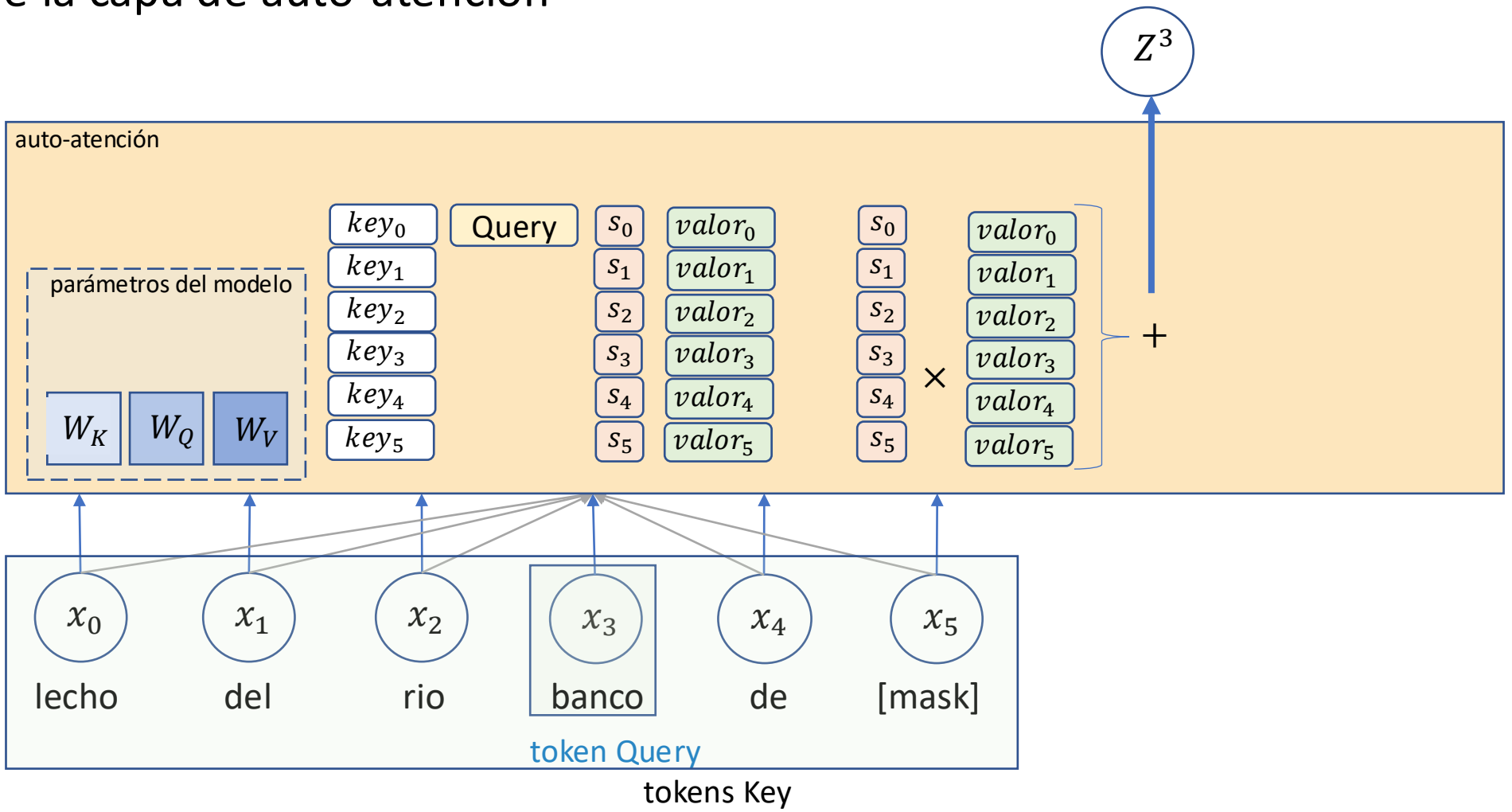
- Los Valores



Embeddings no-Contextual X_i

Transformers

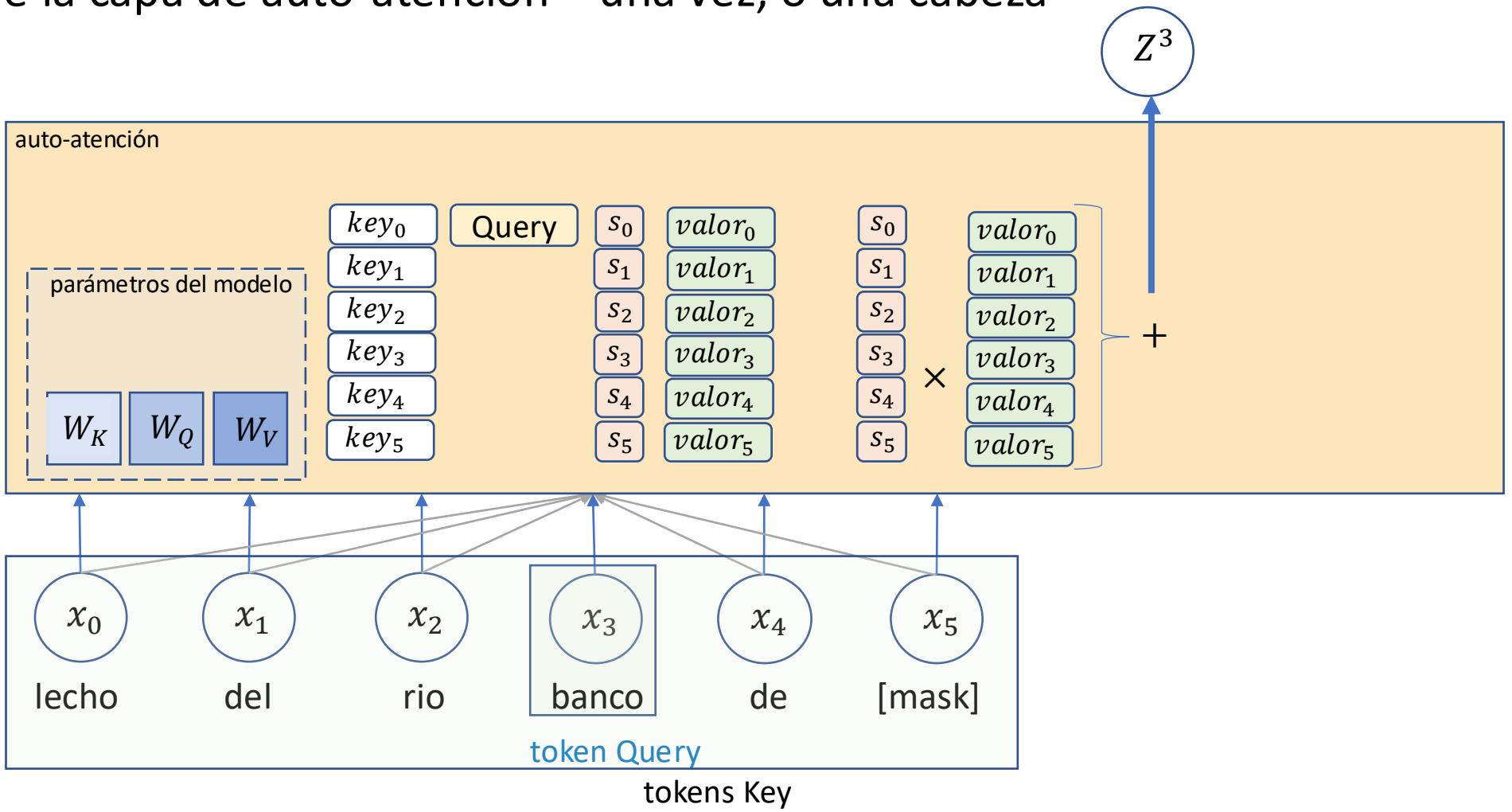
- La Salida de la capa de auto-atención



Embeddings no-Contextual X_i

Transformers

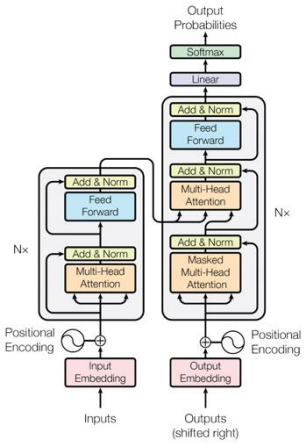
- La Salida de la capa de auto-atención – una vez, o una cabeza

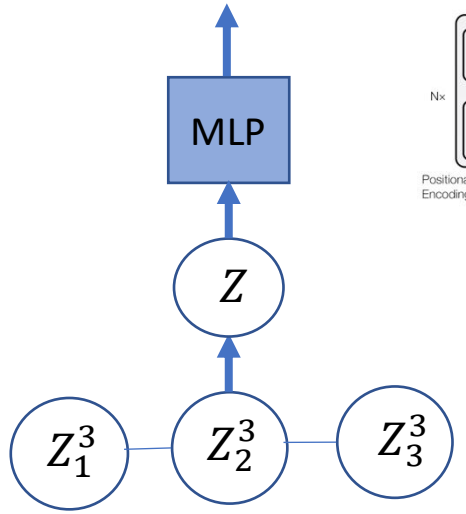


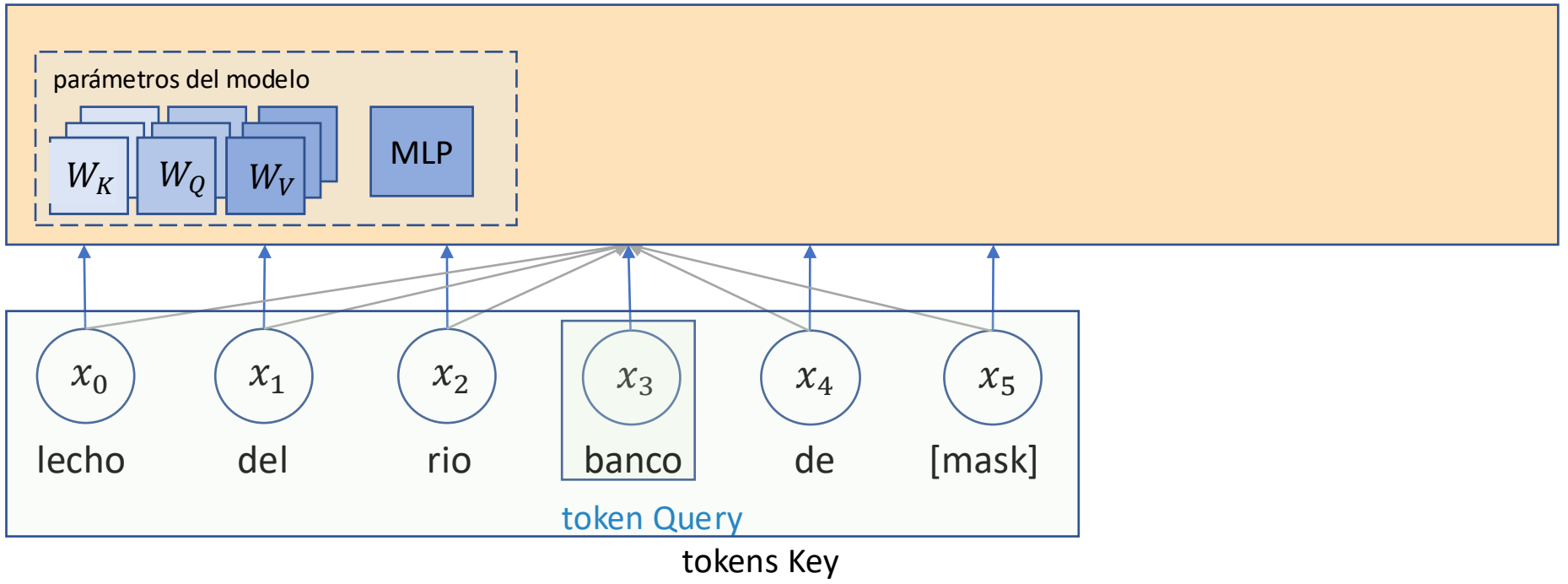
Embeddings no-Contextual X_i

Transformers

- Auto-atención con múltiples cabezas (8 en el transformer)



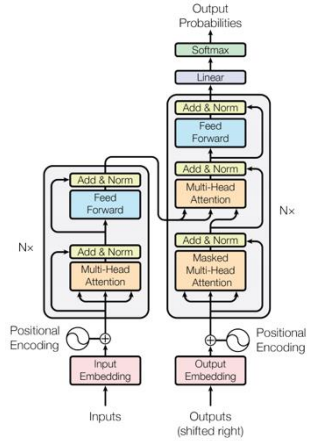




Embeddings no-Contextual X_i

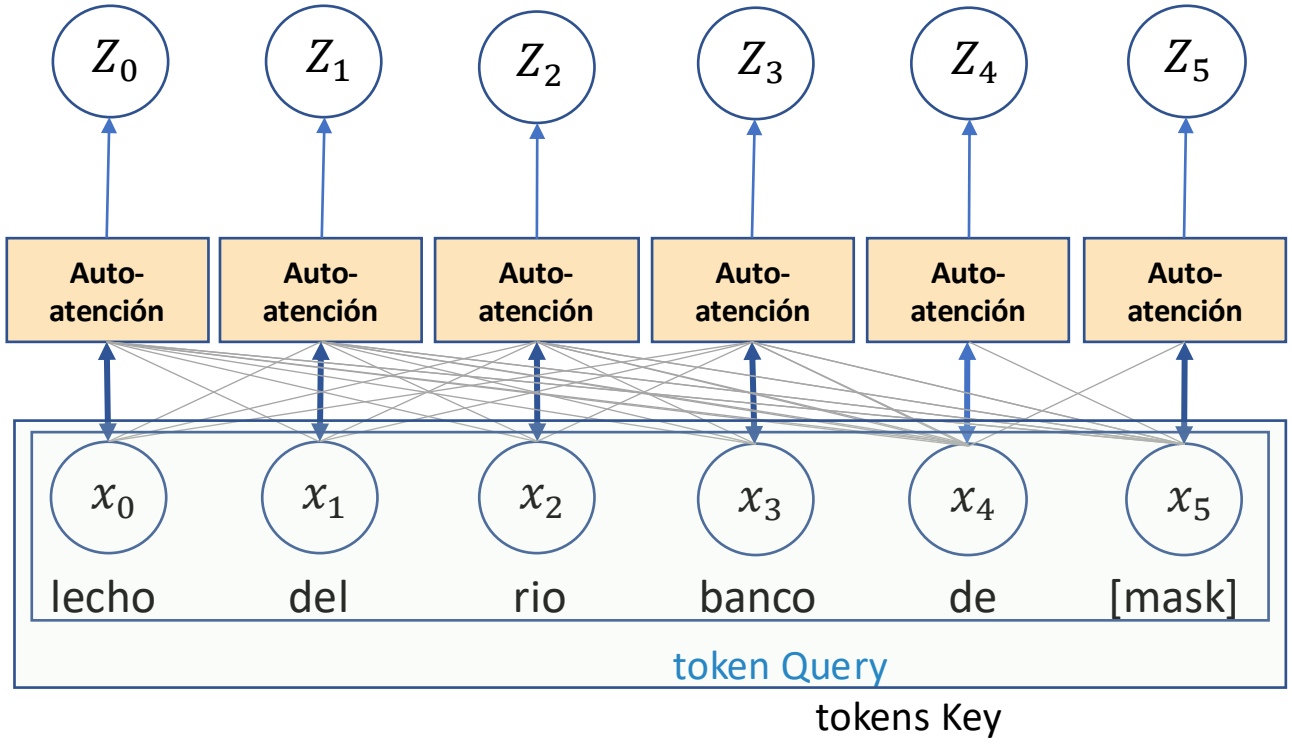
Transformers

- En la salida de las capas de auto-atención, tenemos:



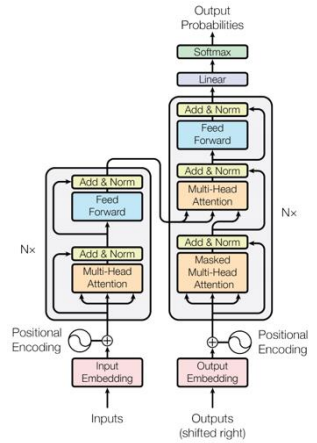
Embeddings Contextual Z_i

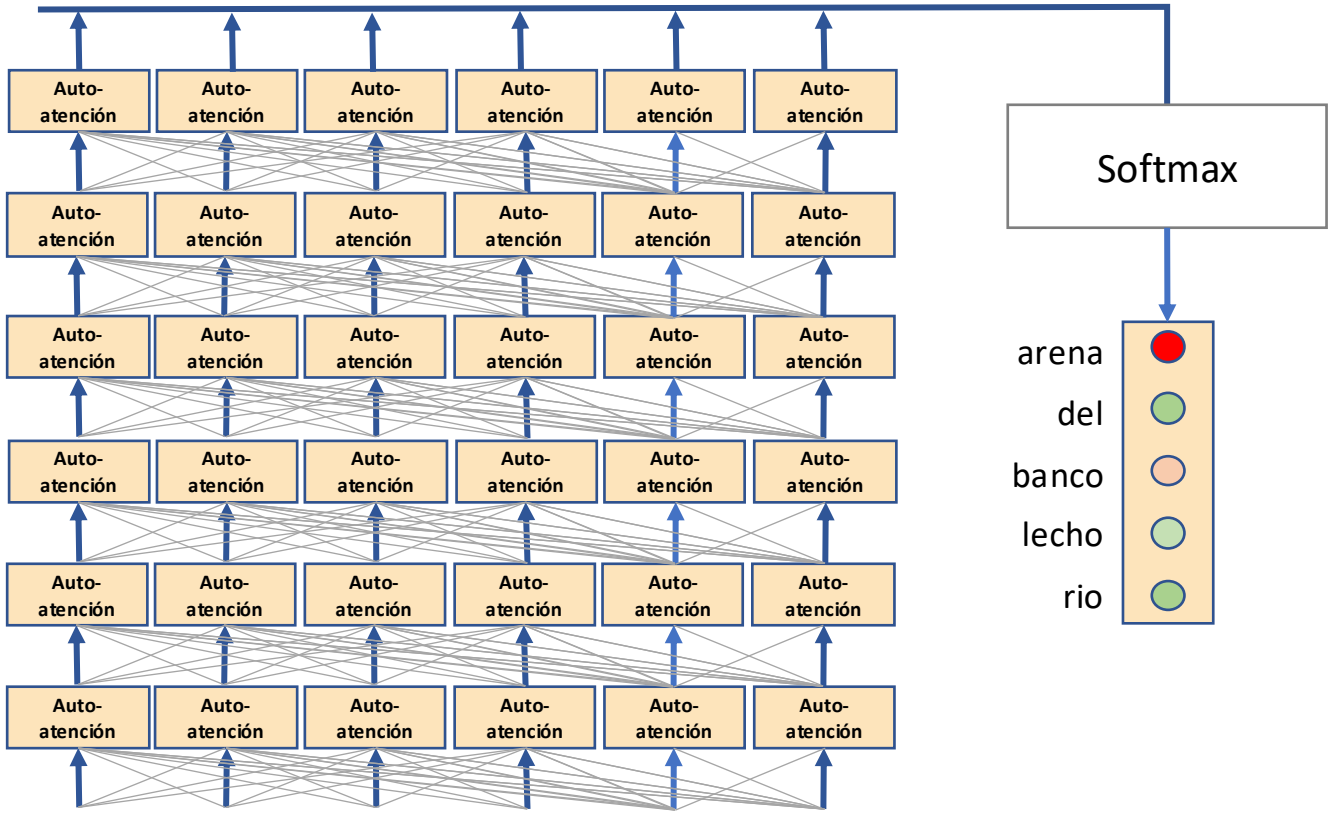
Embeddings no-Contextual X_i



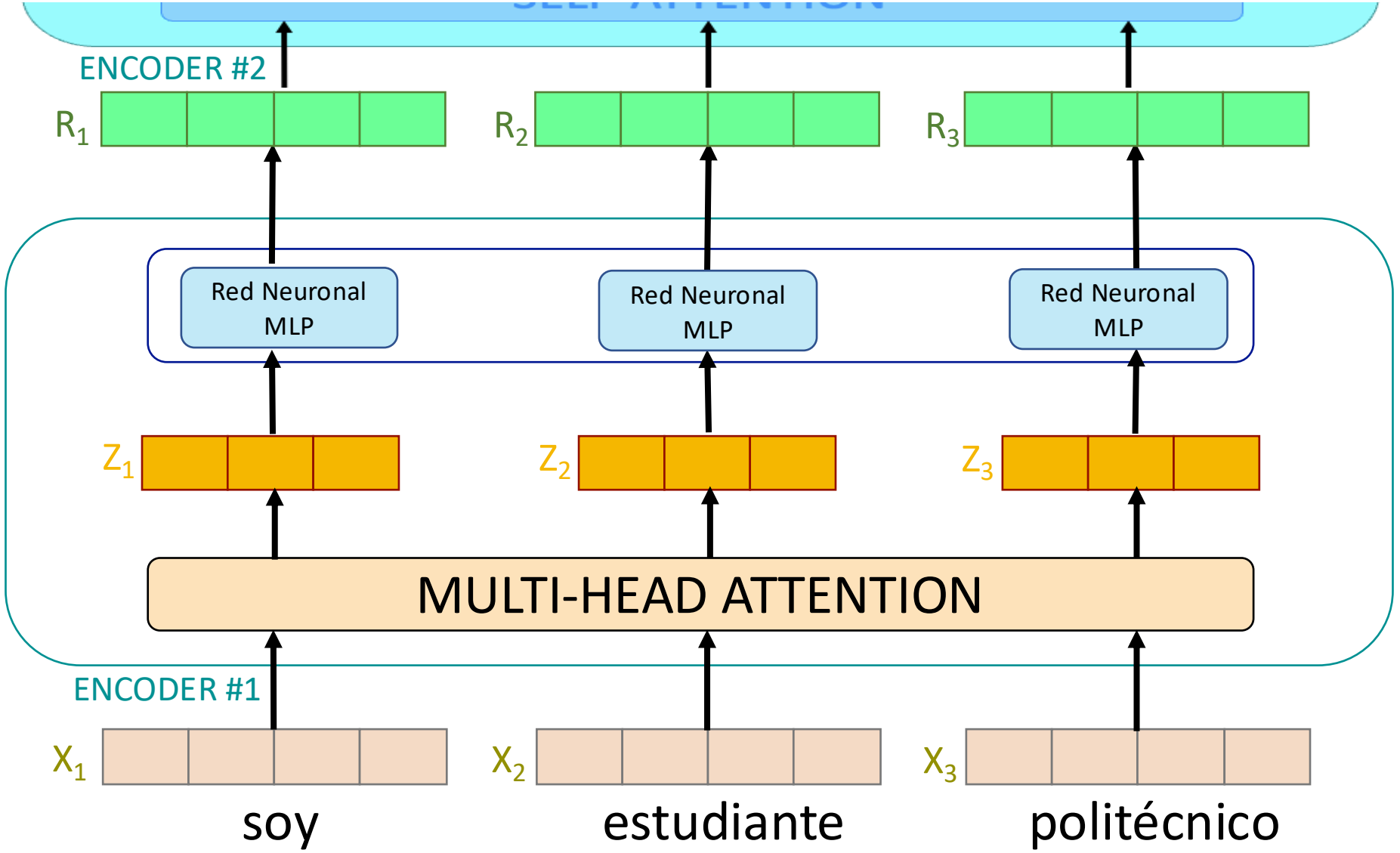
Transformers

- El codificador – Transformer
  - Cabeza o torre de codificación extrae las características del contexto que se pueden clasificar (ViT)

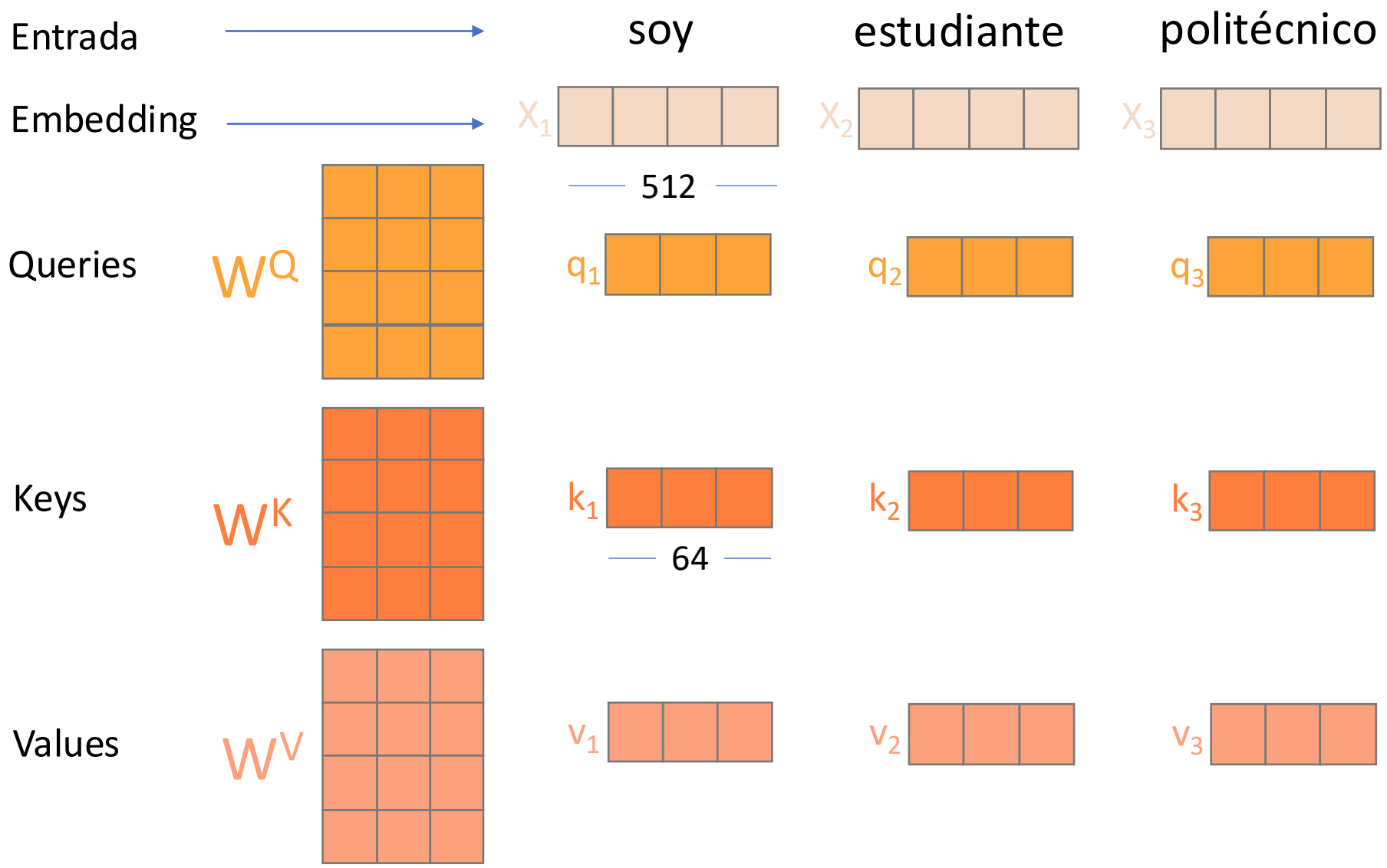




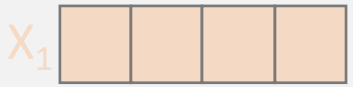
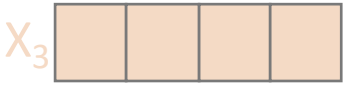
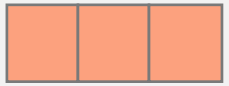
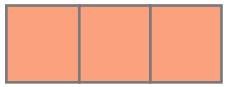
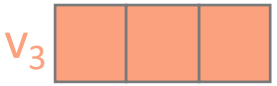
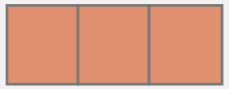
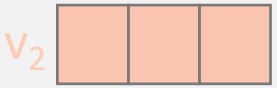
Transformers



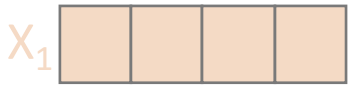
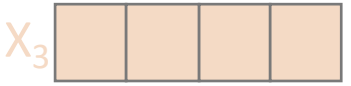
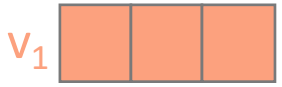
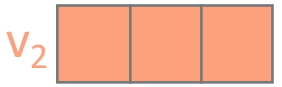
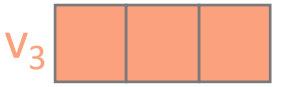
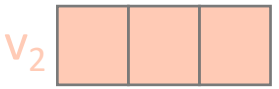
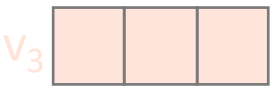
Transformers



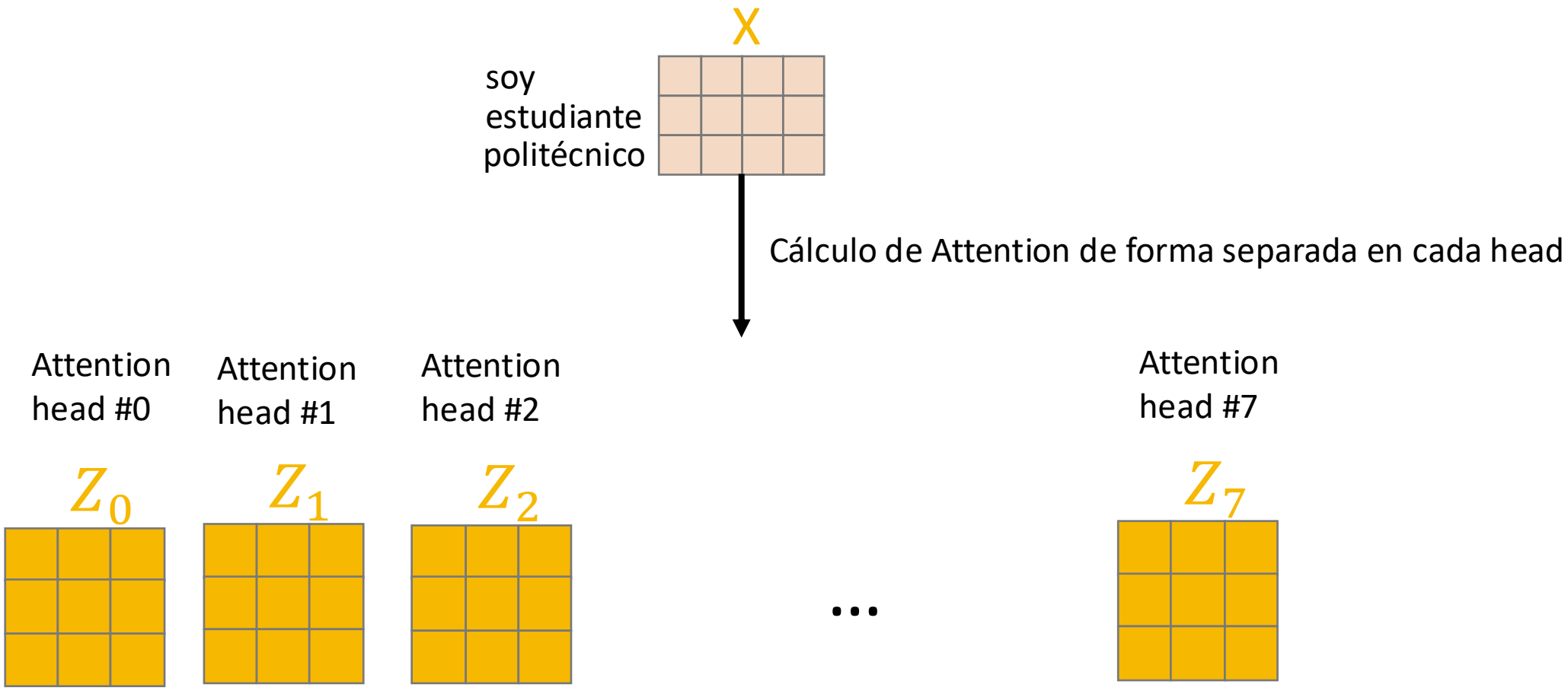
Transformers

	soy	estudiante	politécnico
Entrada			
Embedding			
Queries			
Keys			
Values			
Similitud	$q_1 \cdot k_1^T = 120$	$q_1 \cdot k_2^T = 82$	$q_1 \cdot k_3^T = 61$
Dividir $(\sqrt{d_k} = 8)$	15	10,3	7,6
Softmax	0,74	0,16	0,10
Softmax × Values			
Suma			

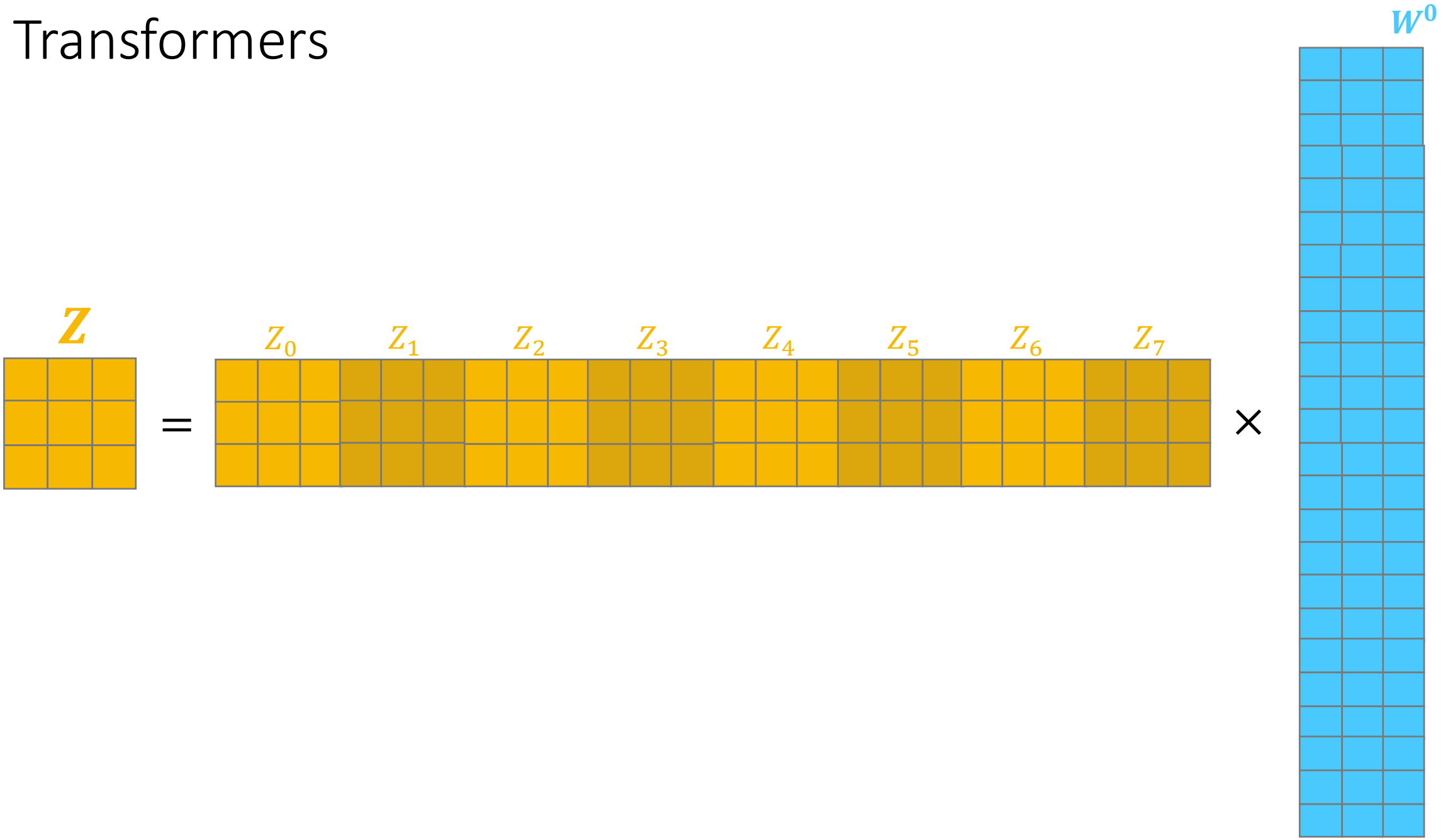
Transformers

Entrada	soy	estudiante	politécnico
Embedding			
Queries			
Keys			
Values			
Similitud	$q_1 \cdot k_1^T = \#\#$	$q_1 \cdot k_2^T = \#\#$	$q_1 \cdot k_3^T = \#\#$
Dividir $(\sqrt{d_k} = 8)$	##	##	##
Softmax	##	##	##
Softmax × Values			
Suma			

Transformers



Transformers



Transformers

